Arrive Home BI & Data Architecture Overview

1. Overview

Objective

Build a centralized Business Intelligence (BI) platform to unify all data pipelines, automate reporting, and replace legacy Homebase reports.

This initiative supports Arrive Home's lending, servicing, and investor operations, ensuring full visibility across Down Payment Assistance (DPA), White Label, and Earned Equity Program (EEP) business lines.

Scope

The BI system will integrate data from all core operational systems, standardize it within a controlled data model, and deliver accurate, timely Power BI dashboards for operations, compliance, finance, and executive stakeholders.

2. Company Context

Arrive Home Overview

Arrive Home is the market leader in Down Payment Assistance (DPA).

Monthly loan volume averages approximately **$0.5 billion**, with about **65 percent (or $325 million)** in funded loans each month.

All newly processed loans are being transitioned through Homebase by the end of the year.

EEP workflows are currently managed partly through the CS Portal and manual spreadsheets, and the development team's goal is to consolidate these processes within Homebase.

3. Data Pipelines

System	Purpose	Integration Requirement	Data Handling / Notes
Homebase Loan Management System (HB)	Core platform managing all lending workflows.	Central hub for data; MongoDB backend with tightly controlled schema.	All existing reports currently reside here; all business data flows through HB.
LendingPad	Loan Origination System for refinancing.	Two-way integration with Homebase.	Enables push and pull of refinance loan data.
VisionNet	Data quality and managed services provider.	Reporting on data-quality checks against Homebase.	Provides spot-check data and exception feeds for BI dashboards.
Data Tapes (Securitized Loans)	Spreadsheets representing securitized loan pools.	Uploaded or emailed to Homebase.	BI must extract and standardize key metrics for investor and financial reporting.
BSI	Servicer for second mortgages.	Manual blob-file upload into Homebase.	BI will ingest, normalize, and report on servicing performance.
ServBank	Servicer for first mortgages.	Pull integration into Homebase.	Provides servicing, payment, and delinquency data.
Black Lake	Document ingestion and indexing platform.	Produces structured metadata and parsed loan-document fields.	Serves as an independent integration point for BI pipelines, reducing dependence on Homebase.

4. Proposed Approach

The goal of this initiative is to establish a unified BI environment that connects all critical lending and servicing data sources to Homebase while maintaining architectural independence from it.

The BI platform will act as a reliable, self-contained reporting layer that remains functional even if Homebase experiences outages or schema changes.

Key Focus Areas

- **Data Integration**
    - Connect Homebase, LendingPad, VisionNet, BSI, ServBank, and securitized loan data tapes into a consolidated reporting foundation.
    - Ingest structured data outputs from **Black Lake** following document parsing and indexing.
    - Evaluate the possibility of capturing or "intercepting" structured data before it enters Homebase when technically feasible.
    - Prioritize resilience by creating integration points independent of Homebase's internal database.
- **Data Quality and Governance**

Implement validation and tracking processes that operate independently of Homebase to ensure accuracy and consistency across all systems.

- **Reporting and Dashboards**

Rebuild existing Homebase reports in Power BI and expand coverage to include program-level, operational, and executive visibility.

- **Scalability and Independence**

Design integration pipelines and storage architecture that minimize dependencies on any single vendor or schema.

- **Collaboration**

The development team will determine detailed phases, sequencing, and resourcing based on available APIs, data accessibility, and technical assessment.

5. Key Considerations

- **Integration Independence**

While Homebase remains the operational system of record, BI integrations should not depend on direct database access. Prefer stable integration points—through Black Lake or upstream feeds—to ensure long-term reliability.

- **Black Lake Ingestion**

A significant portion of structured loan data will originate from Black Lake's document-processing output. These files can serve as primary input for BI pipelines.

- **System Resilience**

The BI environment should continue functioning even if Homebase is unavailable or its schema changes.

- **Data Flow Discovery**

Early discovery work is required to determine the most effective capture points for data ingestion, validation, and transformation.

- **Manual Processes**

Identify and document manual workflows, particularly within EEP, to guide future integration efforts.

- **VisionNet Coordination**

Establish clear procedures for delivering data-quality results and exception reporting.

- **Servicing Feeds**

Define schemas and secure exchange methods for BSI and ServBank data.

- **Governance**

Implement version control for schema updates and maintain consistent report validation practices.

6. Success Criteria

- All lending and servicing data centralized within the BI warehouse.
- All Homebase reports replaced with Power BI dashboards.
- Automated data-refresh cycles with less than 1 percent variance across systems.
- Manual reporting and reconciliation reduced by at least 80 percent.
- Data lineage and ownership documented across all pipelines.

7. Risks and Mitigation

Risk	**Impact**	**Mitigation**
Integration complexity (ServBank, BSI)	High	Define data contracts and test feeds early.
Transition to Black Lake	Medium	Validate metadata formats before production integration.
Incomplete source documentation	High	Conduct discovery and schema-mapping sessions.
Data-quality variance	High	Implement validation and exception tracking through VisionNet.
Resource bottlenecks	Medium	Maintain flexible contracting model and shared oversight.

9. Next Steps

- Confirm access and schema documentation for all relevant systems.
- Align with the development team on sequencing and resource allocation.
- Approve this project definition and initiate data-discovery and design work.
- Establish a recurring BI coordination meeting between Arrive Home and Ascent Factory.

- Define data-governance standards for report maintenance and validation.

10. Loan Lifecycle and Data Flow Overview

The following outlines how a loan progresses through Arrive Home's systems and where data originates for BI reporting.

Down Payment Assistance (DPA) Loans

1. **Loan Registration**
    a. Correspondent lenders (Arrive Home's clients) start the loan process and register the loan with **Homebase** by submitting a **MISMO XML** file.
2. **Funding Obligation Letter**
    a. Arrive Home returns a **funding obligation letter**, confirming commitment to fund the second mortgage. This letter allows the correspondent to close the loan.
3. **Rate Lock and Confirmation**
    a. Before closing, lenders may lock the first mortgage rate, typically between **101–103 percent**. Arrive Home sends a **lock confirmation**.
4. **Loan Closing and Package Delivery**
    a. After closing, the correspondent sends a **complete loan package**—often a **1,000-page PDF**—to Arrive Home.
    b. **Black Lake** processes this package, parsing it into indexed documents and structured metadata. Missing or incomplete documentation triggers conditions that are sent back to the correspondent for correction.
5. **Purchase and Warehouse Processing**
    a. Once conditions are cleared, Arrive Home purchases the loan.
    b. Loans are grouped into **warehouse batches** funded through **South Star Bank**, Arrive Home's warehouse line partner.
6. **Servicing and Ownership**
    a. The **first mortgage** is serviced by **ServBank**; the **second mortgage** is serviced by **BSI**.
    b. Arrive Home may hold loans or sell them; defaults are repurchased by the original correspondent.
7. **Loan Features and Securitization**

a. The second mortgage is typically **2 percent above the first**, with a **10-year term vs. 30 years** on the first.
b. Borrowers locked in for three years receive forgiveness on the second if payments are made for the full period.
c. Loans may later be **securitized** and sold to investors such as **United Security Financial (USF)** or through **Ginnie Mae** programs managed via **Mountain West**.

White Label Programs

- Designed for larger correspondents where Arrive Home only purchases the **second mortgage**.
- No warehouse line is involved.
- Reconciliation occurs through **three-year batches**, determining amounts to wire and corresponding interest revenue.

Earned Equity Program (EEP)

- Serves borrowers who cannot qualify for traditional loans.
- Arrive Home (or a partner such as the **Tule River Tribe**) purchases the property, and the borrower enters a **lease-to-own** arrangement.
- Three instruments are involved:
    - First mortgage
    - Second mortgage
    - Lease agreement
- Custom underwriting and spreadsheet-based processes are used today, which are being migrated into Homebase for full integration.